

Arash Jalil Khabbazi, MS

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Profile

I work on reinforcement learning (RL) and model predictive control (MPC) for smart buildings and energy systems. I also explore broader machine learning (ML) methods in other AI-enabled applications

Education

Purdue University, IN, USA PhD in Mechanical Engineering — GPA: 3.9/4.0 Minor in Computational Science and Engineering	2023 – 2027
University of British Columbia, BC, Canada MS in Mechanical Engineering — GPA: 4.0/4.0 (94%)	2021 – 2023
University of Tabriz, EA, Iran BS in Mechanical Engineering — GPA: 4.0/4.0 (19.12/20) — Rank: 1/155+	2016 – 2020

Experience

Applied ML Intern, Tesla, Inc., Sparks, NV, USA • Built supervised ML models (XGBoost, DNNs, SVR, and ensembles) for defect detection and quality assessment. • Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models. • Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing. • Collaborated on projects projected to reduce testing costs by over \$1M annually through improved quality assessment.	05/2025 – 08/2025
PhD Research Assistant, Purdue University, West Lafayette, IN, USA • Developed a Soft Actor-Critic RL HVAC controller in OCHRE-Gym , saving 17.5% energy while maintaining comfort. • Reviewed 100+ publications on MPC and RL in HVAC, leading to a published review paper in Applied Energy. • Formulated a multi-zone HVAC M&V method to remove savings overestimation from cross-zone heat transfer. • Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy). • Supported commissioning of Herrick Labs by upgrading the building automation system and integrating Modbus. • Designed HVAC control strategies with MPC and DeePC, showing DeePC matched MPC without a system model. • Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.	08/2023 – 08/2027
MS Research Assistant, University of British Columbia, BC, Canada • Modeled hydrogen injection into natural gas pipelines with CFD by developing ideal- and real-gas mixture models. • Produced a journal paper , three conference papers, two awards, and a master's thesis based on the research findings.	05/2021 – 08/2023
Student Researcher, FortisBC Energy Inc., Surrey, BC, Canada • Performed assessments on hydrogen injection into natural gas pipelines and renewable gas projects (UBC H ₂ Lab). • Worked with engineers and research teams (materials, sensors, AI, safety) to assess industry impacts.	05/2021 – 08/2023

Skills

- **Programming:** Python, C, MATLAB, R, MySQL, Git, Linux
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Stable-Baselines3, XGBoost
- **Data & Analytics:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Grafana
- **Modeling & Optimization:** CVX, CVXPY
- **Simulation & Control Testbeds:** BOPTEST, EnergyPlus, OpenAI Gym, OCHRE, OCHRE-Gym, TRNSYS
- **Systems & Integration:** Tridium Niagara, InfluxDB, Modbus, IO-Link, MTConnect

Selected Publications — Full list on [Google Scholar](#)

- **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premier, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, K.J. Kircher. “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review.” *Applied Energy*, 2025. [\[DOI\]](#)
- **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, J. Quinn. “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions.” *International Journal of Hydrogen Energy*, 2024. [\[DOI\]](#)
- Six peer-reviewed conference papers (ACM BuildSys, CISBAT, IHPB, CSME, IGEC; 2021–2025).

Selected Honors & Awards

Best PhD Forum Presentation Runner-up Award at ACM BuildSys (2025) — ASHRAE Graduate Student Grant-in-Aid Award (2025) — Best Paper Award at CSME (2023) — Best Presentation Award at CSME (2022) — UBC Graduate Scholarship (2022) — UBC Dean’s Entrance Scholarship (2021)

Selected Courses

Artificial Intelligence (ECE570) — Reinforcement Learning and Control (IE690) — Applied Machine Learning (UBC) — Intro to Convex Optimization (AAE561) — Industrial IoT Implementation (ME597) — Statistical Methods (STAT511) — Distributed Energy Resources (ME597) — Analysis of Thermal Systems (ME511) — Advanced Thermodynamics (ME500) — Advanced Mathematics (MATH527)